Ministry of Communications and Information Technology

NTI National Telecommunication Institute

FINAL PROJECT REPORT

Roaya

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Smart City Transformation

A city that saves your time, your money, and your planet.

Project	Roaya – Smart City Transformation
Duration	24 months · January 2027 – December 2028
Total budget	EGP 550,000,000
Team	Project Manager, Planning Manager, Technical Manager, Finance Manager, Risk & Quality Manager, Stakeholder Manager
Team members	1. Karim Mohamed Gad 2. Mai Nasser Abubakr 3. Mariam Mahmoud Dabeesh 4. Moaz Hamed Ibrahim 5. Omnia Sayed Abdelhakim 6. Rahma Khaled Ahmed
Program	NTI – Final Project (Project Management)
Date	9 November 2026

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rahmaoshba.github.io/RahmaOshba

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1. Executive Summary

Roaya (Arabic for "vision") is a 24-month project to turn a normal city into a smart city. It uses technology to make city life easier, safer and greener, and to give citizens better public services. The project runs from January 2027 to December 2028 with a total budget of EGP 550 million.

The work is planned in 7 phases: assessment, roadmap and business case, architecture and governance design, infrastructure modernization, technology integration and testing, a pilot in one district, and city-wide deployment. Each phase ends with a milestone that the sponsor approves before the next phase starts.

24	EGP 550M	7	6	8
months	total budget	phases	deliverables	milestones

This report brings together the project charter, schedule, resource allocation, budget, risk management, quality assurance and communication plans. Together they show how the team will deliver the project on time, within budget and to the required quality.

2. Project Charter

2.1 Purpose

To use technology to make city life easier, safer and greener, and to give citizens better public services.

2.2 Objectives

#	Objective
1	Reduce traffic jams by 30% within 2 years.
2	Reduce energy and water use by 20% within 2 years.
3	Easy access: reach anywhere in the city in under 15 minutes.
4	Green energy: 50% of the city's power from solar energy.
5	Safe city: emergency response in under 8 minutes.
6	Jobs: create 20,000 jobs.
7	Enhance security systems and surveillance.
8	Make 80% of government services available online by the end of 2028.

2.3 Expected Impact

- Better quality of life for citizens.
- A cleaner environment, with less pollution and waste.
- New jobs and investment in technology.

2.4 Scope

In scope: current state assessment, roadmap and business case, integration architecture, infrastructure upgrades (networks, sensors, cameras, solar energy), the control center and online services, testing, a pilot in one district, and city-wide rollout with staff training.

2.5 Deliverables

#	Deliverable	Due date	Phase
D1	Current State Assessment	31 Mar 2027	Phase 1
D2	Phased Roadmap & Business Case	30 Jun 2027	Phase 2
D3	Integration Architecture	30 Sep 2027	Phase 3
D6	Security, Risk & Data Governance Framework	30 Sep 2027	Phase 3
D4	Pilot Deployment & KPI Report	30 Sep 2028	Phase 6
D5	Scale-up & Change Management Plan	31 Oct 2028	Phase 7

2.6 Stakeholders

Government authorities, project sponsors, technology providers, infrastructure contractors, utility companies and, most importantly, city residents.

2.7 Team Roles

Role	Main responsibility
Project Manager	Leads the project, approves each phase and reports to the sponsor
Planning Manager	Schedule, milestones and the Gantt chart
Technical Manager	Smart city technologies and infrastructure
Finance Manager	Budget, funding and procurement
Risk & Quality Manager	Risks, cybersecurity and quality
Stakeholder Manager	Stakeholders, engagement and communications

3. Project Schedule and Milestones

The project is completed in 24 months (January 2027 – December 2028) using a phased approach. The full task list and Gantt chart are in the separate schedule file.

3.1 Phases

Phase	Name	Period	Main owner
1	Initiation & Current State Assessment	Jan 2027 – Mar 2027	Project Manager
2	Roadmap & Business Case	Apr 2027 – Jun 2027	Finance Manager
3	Architecture & Governance Design	Jul 2027 – Sep 2027	Technical Manager
4	Infrastructure Modernization	Oct 2027 – Mar 2028	Technical Manager
5	Smart Technology Integration & Testing	Apr 2028 – Jun 2028	Technical Manager
6	Pilot Operation	Jul 2028 – Sep 2028	Technical Manager
7	City-wide Deployment & Closure	Oct 2028 – Dec 2028	Project Manager

3.2 Milestones

ID	Milestone	Date	Owner
M1	Deliverable 1: Current State Assessment approved	31 Mar 2027	Project Manager
M2	Deliverable 2: Phased Roadmap & Business Case approved	30 Jun 2027	Project Manager
M3	Deliverables 3 & 6: Integration Architecture and Governance Framework approved	30 Sep 2027	Project Manager
M4	Infrastructure ready	31 Mar 2028	Technical Manager
M5	Systems tested and ready	30 Jun 2028	Risk & Quality Manager
M6	Deliverable 4: Pilot Deployment & KPI Report approved	30 Sep 2028	Project Manager
M7	Deliverable 5: Scale-up & Change Management Plan approved	31 Oct 2028	Project Manager
M8	Project closure	31 Dec 2028	Project Manager

4. Resource Allocation

4.1 Resource Categories

Category	Description
Human Resources	Core team of 6 managers (Project, Planning, Technical, Finance, Risk & Quality, Stakeholder); engineers and technicians; software, data and cybersecurity specialists; contractor crews; trainers and support staff
Material Resources	Fiber cables, construction materials, training manuals, awareness brochures, project templates and reports
Equipment	IoT sensors, surveillance cameras, smart meters, smart traffic signals, solar energy systems, servers, control center workstations, laptops and field vehicles
Financial Resources	EGP 500M main budget + EGP 50M backup budget = EGP 550M; extra funding from investors and private partners

Category	Description
Time	24 months (Jan 2027 – Dec 2028) in 7 phases: assessment, roadmap, design, infrastructure, integration and testing, pilot, city-wide rollout

4.2 Allocation by Task

Task	Responsible	Duration	Key resources	Cost (EGP M)
Current state assessment	Technical Manager	10 weeks (Jan – Mar 2027)	Laptops, survey forms	4
Stakeholder mapping and engagement plan	Stakeholder Manager	7 weeks (Jan – Feb 2027)	Meeting rooms, survey forms	1
Initial risk assessment	Risk & Quality Manager	6 weeks (Feb – Mar 2027)	Risk register template	1
Phased roadmap and business case	Planning Manager + Finance Manager	3 months (Apr – Jun 2027)	Planning software, laptops	4
Integration architecture design	Technical Manager	3 months (Jul – Sep 2027)	Design software, test servers	20
Security and data governance framework	Risk & Quality Manager	3 months (Jul – Sep 2027)	Cybersecurity tools	12
Network and fiber upgrade	Technical Manager	4 months (Oct 2027 – Jan 2028)	Fiber cables, network equipment	80
Install sensors and surveillance cameras	Technical Manager	5 months (Nov 2027 – Mar 2028)	IoT sensors, cameras, field vehicles	100
Install solar energy systems	Technical Manager	6 months (Oct 2027 – Mar 2028)	Solar panels, construction materials	45
Manage contractors and procurement	Finance Manager	6 months (Oct 2027 – Mar 2028)	Contracts, procurement system	13
Control center and online services	Technical Manager	3 months (Apr – Jun 2028)	Servers, workstations, software	100
System and security testing	Risk & Quality Manager	2 months (May – Jun 2028)	Testing tools	20
Pilot in one district	Technical Manager + Stakeholder Manager	3 months (Jul – Sep 2028)	Smart devices, citizen app, feedback forms	43
City-wide rollout and staff training	Technical Manager + Stakeholder Manager	2 months (Oct – Nov 2028)	Smart devices, training manuals, brochures	52
Final review and project report	Project Manager	1 month (Dec 2028)	Laptops, reports	5
Contingency (backup budget)	Project Manager	When needed	–	50
Total				**550**

4.3 Current Availability

Peak team: 186 people in Phase 4 (Infrastructure), 120 of them contractor crews.

Resource	Available now	How we fill the gap
Core project team	6 of 6	Already assigned
Engineers and technicians	5 of 46	Hire and use supplier contracts
Software, data and cybersecurity team	1 of 21	Hire or outsource
Contractor crews	0 of 120	Infrastructure contractors
Cameras and fiber network	Partly (300 cameras, 120 km fiber)	Procure the rest

Resource	Available now	How we fill the gap
Sensors, solar and control center	Not available	Procure through technology and energy partners

5. Budget and Funding

5.1 Budget by Cost Item

Item	Cost (EGP)	% of total
Infrastructure (sensors, networks, cameras)	250,000,000	45.5%
Software, apps and the control center	150,000,000	27.3%
Cybersecurity	60,000,000	10.9%
Training and awareness	40,000,000	7.3%
Subtotal: main cost	**500,000,000**	**90.9%**
Backup budget (about 10%)	50,000,000	9.1%
Total budget	**550,000,000**	**100%**

5.2 Budget by Phase

Phase	Name	Cost (EGP M)	% of total
1	Initiation & Current State Assessment	6	1.1%
2	Roadmap & Business Case	4	0.7%
3	Architecture & Governance Design	32	5.8%
4	Infrastructure Modernization	238	43.3%
5	Smart Technology Integration & Testing	120	21.8%
6	Pilot Operation	43	7.8%
7	City-wide Deployment & Closure	57	10.4%
–	Backup budget (contingency)	50	9.1%
	Total	**550**	**100%**

Phase 4 (infrastructure) uses the largest share of the budget, because it covers networks, sensors, cameras and solar systems.

5.3 Funding and Revenue

- **Funding:** the main government budget, plus extra funding from investors and private partners.
- **Revenue:** selling and renting land, offices and homes; transport tickets and highway tolls; selling extra solar energy to the national grid; renting data center space; and fees for smart services (parking, ads on smart screens).
- **Cost control:** the Finance Manager compares budget vs. actual every month and flags any spending more than 10% over plan.

6. Risk Management

Each risk is scored by multiplying probability (1–5) by impact (1–5). Risks scoring 12 or more are High: they have a contingency plan and are reviewed every week.

ID	Risk	Score	Severity	Strategy	Response · Owner
R1	Cyberattack on city systems	15	**High**	Reduce	Firewalls, 24/7 monitoring, regular security testing · *Risk & Quality Manager*
R2	Technical issues with smart devices	12	**High**	Reduce	Test devices before rollout and keep spare units · *Technical Manager*
R3	Budget overrun	12	**High**	Reduce	Monthly cost review; use the EGP 50M backup budget · *Finance Manager*
R4	Data privacy concerns	12	**High**	Avoid	Clear privacy policy and follow data protection law · *Risk & Quality Manager*
R6	Contractor or supplier delays	9	**Moderate**	Transfer	Contracts with deadlines and late-delivery penalties · *Planning Manager*
R7	Insufficient user adoption	9	**Moderate**	Reduce	Awareness campaigns, easy app, training sessions · *Stakeholder Manager*

ID	Risk	Score	Severity	Strategy	Response · Owner
R5	Delays in regulatory approvals	8	Moderate	Reduce	Start approvals early and assign one person to follow up · *Project Manager*

6.1 Contingency Plans for High Risks

Risk	Trigger	Emergency action
Cyberattack on city systems	An attack or data breach is detected	Isolate the affected systems, activate the incident response team, inform authorities and citizens
Technical issues with smart devices	More than 5% of devices fail in one area	Replace with spare units and switch to manual operation until fixed
Budget overrun	Spending is more than 10% over plan	Release the backup budget and delay low-priority work
Data privacy concerns	Public complaints or legal questions	Pause the related service, review the privacy policy and publish a clear statement

6.2 Monitoring

- **Weekly:** high risks are reviewed in the team meeting.
- **Monthly:** the Risk & Quality Manager updates the risk register.
- **End of each phase:** a full risk review to find new risks for the next phase.

7. Quality Assurance

7.1 Quality Objectives

- All smart systems work reliably, with at least 99% uptime after launch.
- All equipment meets safety, electrical and cybersecurity standards.
- Citizen data is protected and follows data protection law.
- At least 85% citizen satisfaction during the pilot.
- The project is completed on time (December 2028) and within the EGP 550M budget.

7.2 Quality Assurance Activities

Activity	Purpose	Timing	Responsible
Design review	Check the architecture and governance design meet standards	Phase 3	Technical Manager
Supplier evaluation	Verify the quality and reliability of suppliers	Before purchase	Finance Manager
Installation inspection	Check networks, sensors and cameras are installed correctly	Phase 4	Technical Manager
System and security testing	Test the control center, app and cybersecurity	Phase 5	Risk & Quality Manager
Citizen feedback	Collect opinions on the new services	Pilot and after	Stakeholder Manager

7.3 Quality Control Process

1. Use inspection checklists for design, installation and systems.
2. Test all equipment and systems before they go live.
3. Record any issue in the Quality Log and fix it immediately.
4. Run a final test before the pilot and before the city-wide launch.
5. The Project Manager gives final approval once all criteria are met.

8. Communication Plan

Every update has one owner, a fixed time and a clear channel.

Who	What	When	Audience	How
Project Manager	Progress report and phase approvals	Monthly, end of phase	Sponsor, government	Email + review meeting

Who	What	When	Audience	How
Planning Manager	Status report and Gantt update	Weekly	Core team	Team meeting
Technical Manager	Technical progress report	Every 2 weeks	Contractors, providers	Site meeting, video call
Finance Manager	Budget vs. actual report	Monthly	Sponsor, investors	Email report
Risk & Quality Manager	Risk and security report	Monthly	Core team, sponsor	Team meeting + email
Stakeholder Manager	Citizen updates and service notices	Monthly, before any works	Residents, utilities	Social media, city app

8.1 Escalation

- **Team issues:** solved in the weekly team meeting.
- **Delays, budget or high risks:** sent to the Project Manager within 2 working days.
- **Critical issues (security incident, major delay):** the sponsor is informed within 24 hours.

9. Success Measures (KPIs)

Baseline values are measured in Phase 1 and the KPIs are reported in the Pilot KPI Report (D4) and at project closure.

KPI	Target	Measured	Owner
Traffic congestion	30% less	Pilot and closure	Planning Manager
Energy and water use	20% less	Pilot and closure	Technical Manager
Solar share of city power	50%	Closure	Technical Manager
Emergency response time	Under 8 minutes	Monthly after pilot	Technical Manager
Government services online	80% by end of 2028	Quarterly	Technical Manager
New jobs created	20,000	Closure	Finance Manager
System uptime	99% or more	Monthly after launch	Risk & Quality Manager
Citizen satisfaction	85% or more	Pilot survey	Stakeholder Manager
Schedule and budget	Dec 2028, EGP 550M	Monthly	Project Manager

10. Conclusion and Recommendations

Roaya gives a normal city a new vision: smart, green and connected. With a clear charter, a 24-month phased schedule, a fully allocated EGP 550M budget, and plans for risk, quality and communication, the team is ready to deliver the project on time, within budget and to a high standard.

Recommendations

- Start regulatory approvals and funding talks early, in Phases 1 and 2.
- Treat cybersecurity and data privacy as a priority from the design phase.
- Use the pilot to learn and fix problems before the city-wide rollout.
- Keep citizens informed and involved, so they use the new services.
- Record lessons learned at the end of each phase for future smart city projects.

11. Approval

Name / Role	Signature	Date
Project Sponsor:		
Project Manager:		